

How to use these token cutouts



These pages contain the text *Green Eggs and Ham* by Dr Seuss. Each **cutout** shows a **next word** paired with the **previous word** that came immediately before it in the original text.

Model vitals		
993 tokens	56 unique tokens	1.8 bits/token entropy
3.5 perplexity	3.45 branching factor	

Anatomy of a cutout



Every distinct word has its own colour. **Previous** words appear inside a coloured box (the word's own colour as the background, with contrasting text); the free-standing **next word** appears in plain coloured text. The same word always wears the same colour, whether you see it inside a box or free-standing. Two unrelated words can occasionally share a colour, so always verify the word itself matches — not just the colour.

Setup

Cut out the tokens along the dotted lines and **spread them out** face-up on the table, with no overlap if possible. This is the “training” step: every (previous, next) combination from the original text is now a physical cutout sitting on your table.

How to play: the matching game

To generate text, repeatedly find a cutout whose **previous word** matches the last word you've written, then **write its next word** onto your page. The word you just wrote becomes part of what you'll match against next — that's the chain.

Worked example

Step 1. Pick any cutout to begin — say this one — and copy its previous word *and* its next word onto your page.

Step 2. Look at the last word you've written (**Sam**) and find a cutout whose previous word matches it. Write its next word.

Step 3. Repeat: find a cutout whose previous word matches the last word you've written, and add its next word.

Keep going — write as much or as little as you like.

the cutout

that Sam

Sam I

I am

your page

that Sam

that Sam I

that Sam I am

Tips

Use colour as a fast filter. Scan by the rightmost previous-word box's colour first, then verify the actual word matches.

Pick from many candidates by eye. The more cutouts share the same previous word, the more often your eye lands on one — that's **weighted sampling** for free, and it's why some words follow others more often in your text.

Put the cutout back after using it — removing it would change the model's distribution next time round.

I I am am Sam Sam . I I am am Sam

Sam . Sam Sam I I am am . that

that Sam Sam I I am am ! ! that that Sam

Sam I I am am ! ! I I do do not not like

like that that Sam Sam I I am am ! ! would

would you you like like green green eggs

eggs and and ham ham ? ? I I do do not

not like like them them , , Sam Sam I I am

am . I I do do not not like like green

green eggs eggs and and ham ham . would

eggs green and eggs ham and ham would

like green not like do not I do I am

I am I Sam I, Sam like them not like

and eggs ham and ? ham I ? do I not do

you would like you like green eggs

that I like that Sam I am I am I would

I am I am I do I do not like

• Sam Sam I am • that •

The diagram illustrates the construction of the sentence "I am Sam" using a sequence of tokens. The tokens are shown in a sequence of boxes, each with a different color and font style. The tokens are: "I" (grey, serif), "am" (green, sans-serif), "Sam" (orange, serif), and a period "." (blue, sans-serif). The final sentence "I am Sam" is shown in a single line, with each word in a different color and font style, matching the tokens in the sequence.

would you you like like them them here

here or or there there ? ? I I would

would not not like like them them here

here or or there there . . I I would

would not not like like them them anywhere

anywhere . . I I do do not not like

like green green eggs eggs and and ham

ham . . I I do do not not like like them

them , , Sam Sam I I am am . . would

would you you like like them them in in a

in a	like them in	like them	you like	would you
I would	am	I am	Sam I	Sam
like them	not like	do not	I do	ham
and ham	eggs and	green	eggs	like green
not like	do not	I do	I	anywhere
them anywhere	like them	not like	would not	
I would	I	there	or there	here or
them here	like them	not like	would not	
I would	I	there ?	or there	here or
them here	like them	you like	would you	

a house house ? ? would would you you like

like them them with with a a mouse

mouse ? ? I I do do not not like like them

them in in a a house house . . I I do do not

not like like them them with with a

a mouse mouse . . I I do do not not like

like them them here here or or there there .

. I I do do not not like like them

them anywhere anywhere . . I I do do not

not like like green green eggs eggs and

you like	would you	? would	house ?	a house
a mouse	with a	them with	like them	
like them	not like	do not	I do	? I mouse ?
do not	I do	house	a house	in a
in	them in			
with a	them with	like them	not like	
not like	do not	I do	mouse	
there	or there	here or	them here	like them
like them	do not	I do	do not	I do
do not	I do	anywhere	them anywhere	
eggs and	green eggs	like green	not like	

and ham ham . I I do do not not like

like them them , , Sam Sam I I am am .

. would would you you eat eat them

them in in a a box box ? ? would

would you you eat eat them them with

with a a fox fox ? ? not not in in a a box

box . . not not with with a a fox fox . . not

not in in a a house house . . not not with

with a a mouse mouse . I I would

would not not eat eat them them here

here or or there there . . I I would

would not not eat eat them them anywhere

anywhere . . I I would would not not eat

eat green green eggs eggs and and ham

ham . . I I do do not not like like them

them , , Sam Sam I I am am . . would

would you you ? ? could could you you ?

? in in a a car car ? ? eat eat them them !

! eat eat them them ! ! here here they

they are are . . I I would would not not ,

not,	would	not	I	would	I	are	they	are
here	they	here	them	eat	eat	them	eat	eat
them	eat	eat	them	eat	car	a	in	a
you	could	you	could	you	would	you	would	you
would	am	I	am	Sam	I	Sam	them	,
like	them	not	like	do	not	I	do	I
and	ham	eggs	and	green	eggs	eat	green	
not	eat	would	not	I	would	I	anywhere	
them	anywhere	eat	them	not	eat	would	not	
I	would	I	there	or	there	or	here	or

, could could not not , in in a a car car .

. you you may may like like them them .

. you you will will see see . . you you may

may like like them them in in a a tree

tree ! ! I I would would not not , , could

could not not in in a a tree tree . . not

not in in a a car car ! ! you you let let me

me be be . . I I do do not not like

like them them in in a a box box . . I I do

do not not like like them them with with a

car .	a	car	in	a	, in	not ,	could	you
them .	like	may	like	them	may	like	you	may
you	may	you	will	see	you	will	see	you
a tree	in	a	tree	in	in	like	them	may
, could	not ,	would	not	I	would	I	tree !	
. not	tree .	a	tree	in	a	not in	could	not
let me	you	let	! you	car !	a	car	in	in
not like	do	not	I	do	I	be .	me	be
I do	I	box .	a	box	in	a	like	them
with a	them	with	like	them	not	like	do	not

a fox fox . I I do do not not like

like them them in in a a house house . I

I do do not not like like them them with

with a a mouse mouse . I I do do not

not like like them them here here or

or there there . I I do do not not like

like them them anywhere anywhere . I

I do do not not like like green green eggs

eggs and and ham ham . I I do do not

not like like them them , , Sam Sam I I am

I am	Sam I	, Sam	them,	like them	not like
do not	I do	I	ham.	and ham	eggs and
green eggs	like green	not like	do not	I do	
I	anywhere.	them anywhere	like them		
not like	do not	I do	I	there.	or there
here or	them here	like them	not like		
do not	I do	I	mouse.	a mouse	with a
them with	like them	not like	do not	I do	
I	house.	a house	in a	them in	like them
not like	do not	I do	I	fox.	a fox

am . a a train train ! ! a a train train ! ! a

a train train ! ! a a train train ! ! could

could you you , , would would you you on

on a a train train ? ? not not on on a

a train train ! ! not not in in a a tree tree !

! not not in in a a car car ! ! Sam Sam !

! let let me me be be ! ! I I would

would not not , , could could not not , , in

in a a box box . . I I would would not not ,

, could could not not , , with with a a fox

a fox with a not with could not

a in a box. I would not,

you would 'not' could 'not' in

I would like to meet you

Sam : car : a car in a not in not

tree in a not in not train ! a train !

on a train ? not on a train ? not on a

noʻp
pinoʻ
, noʻ
pinoʻ
, noʻ
pinoʻ
u noʻ

could train a train a train

fox . I I will will not not eat eat them
them in in a a house house . I I will
will not not eat eat them them with with a
a mouse mouse . I I will will not not eat
eat them them here here or or there there .
. I I will will not not eat eat them
them anywhere anywhere . I I do do not
not eat eat green green eggs eggs and
and ham ham . I I do do not not like
like them them , , Sam Sam I I am am .

am . I am Sam I , Sam like them

not like do not I do I ham . and ham

eggs and green eggs eat green not eat

do not I do I anywhere . them anywhere

eat them not eat will not I will I . I

there . or there here or them here eat them

not eat will not I will I mouse . a mouse

with a them with eat them not eat will not

I will I . I house . a house in a them in

eat them not eat will not I will I . I fox .

. say say ! ! in in the the dark dark ?

? here here in in the the dark dark !

! would would you you , , could could you

you , , in in the the dark dark ? ? I

I would would not not , , could could not

not , , in in the the dark dark . . would

would you you , , could could you you , , in

in the the rain rain ? ? I I would would not

not , , could could not not in in the

the rain rain . . not not in in the the dark

the dark	in the	not in	• not	rain •	the rain
in the	not in	could not	could	not	not
would not	I would	I ?	rain ?	the rain	in the
in you	could you	could you	you	would you	you
• would	dark •	the dark	in the	in	not
could not	could	not	would not	I would	
I ?	dark ?	the dark	in the	in you	you
could you	could you	you	would you	!	would
dark !	the dark	in the	here in	?	here
dark ?	the dark	in the	! in	say !	• say

dark . not not on on a a train train . not

not in in a a car car . not not in in a

a tree tree . I I do do not not like

like them them , , Sam Sam , , you you see

see . not not in in a a house house . not

not in in a a box box . not not with

with a a mouse mouse . not not with

with a a fox fox . I I will will not not eat

eat them them here here or or there there .

I I do do not not like like them

• not	a train	on a	not on	• not	dark	•
in a	not in	• not	a car	in a	not in	
not like	do not	I do	• I	tree	a tree	
you see	, you	Sam,	Sam	them,	like	them
• not	house	a house	in a	not in	• not	see
not with	• not	box	a box	in a	not in	
not eat	will not	• I	fox	a fox	with a	
there	or there	here or	them here	eat them		
like them	not like	do not	I do			

them anywhere anywhere ! ! you you do

do not not like like green green eggs

eggs and and ham ham ? ? I I do do not

not like like them them , , Sam Sam I I am

am . . could could you you , , would

would you you , , with with a a goat goat ?

? I I would would not not , , could

could not not with with a a goat goat !

! would would you you , , could could you

you , , on on a a boat boat ? ? I I could

you do ! you anywhere ! them

green eggs like green not like do not

do not I do ? I ham ? and ham and eggs

I am Sam I , Sam them , like them not like

, would you , could you . am .

goat ? a goat with a , with you , would you

, could not , would not I would I ? I

goat ! a goat with a not with could not

could you , could you , would ! would

I could ? I boat ? a boat on a , on you ,

could not not , , would would not not , , on

on a a boat boat . . I I will will not not ,

, will will not not , , with with a a goat

goat . . I I will will not not eat eat them

them in in the the rain rain . . not not in

in the the dark dark ! ! not not in in a

a tree tree ! ! not not in in a a car car !

! you you let let me me be be ! ! I I do

do not not like like them them in in a

a box box . . I I do do not not like

, on , not , would not , would not , could not

not , will not I will I boat . a boat on a

a goat with a , with not , will not , will

eat them not eat will not I will I goat .

not in . not rain . the rain in the them in

in a not in ! not ! dark ! the dark in the

car ! a car in a not in ! not ! tree ! a tree

I do ! I ! be ! me be let me you let ! you

in a in like them in like not like do not

not like do not I do I I box . a box

like them them with with a a fox fox . I

I will will not not eat eat them them in

in a a house house . I I do do not not like

like them them with with a a mouse

mouse . I I do do not not like like them

them here here or or there there . I I do

do not not like like them them anywhere

anywhere . I I do do not not like

like green green eggs eggs and and ham

ham ! ! I I do do not not like like them

like them	not like	do not	I do	I !	ham !
and ham	eggs and	green eggs	like green		
not like	do not	I do	I .	anywhere .	
them anywhere	like them	not like	do not		
I do . I	there .	or there	here or	them here	
like them	not like	do not	I do	I .	mouse .
a mouse	with a	them with	like them		
not like	do not	I do . I	house .	a house	in a
them in	eat them	not eat	will not	I will	
I .	a fox .	with a	them with	like them	

them , , Sam Sam I I am am . . you you do

do not not like like them them . . so so you

you say say . . try try them them ! ! try

try them them ! ! and and you you may

may . . try try them them and and you

you may may , , I I say say . . Sam Sam !

! If If you you let let me me be be , , I

I will will try try them them . . you

you will will see see and and he

he tries tries them them say say !

say !	say	.	.	.	them	tries them	he tries
and he	and	.	.	.	see	will see	you will
	you	.	them	.	try them	will try	I will
, I	be ,	me be	let me	you let	If you	If	If
Sam !	Sam	.	say	.	I say	, I may	you may
and you	them and	try them	.	try	may		
you may	and you	!	and	!	them !	try them	
!	try them !	try them	.	try	say	you say	
so you	so	.	them	.	like them	not like	do not
you do	you	.	am	.	I am	Sam I	Sam ,

! I I like like green green eggs eggs and

and ham ham ! ! I I do do ! ! I I like

like them them , , Sam Sam I I am am !

! and and I I would would eat eat them

them in in a a boat boat . . and and I

I would would eat eat them them with

with a a goat goat . . and and I I will

will eat eat them them in in the the rain

rain . . and and in in the the dark dark .

. and and on on a a train train . . and

eggs and	green eggs	like green	I like	I
I like	I do	I ham	and ham	
am	I am	Sam I	Sam	them
eat them	would eat	I would	and I	and
and I	boat	a boat	in a	them in
them with	eat them	would eat	I would	
I will	and I	and	goat	a goat with a
the rain	in the	them in	eat them	will eat
dark	the dark	in the	and in	and rain
and	train	a train	on a	and on
				and

and in in a a car car . . and and in in a

a tree tree . . they they are are so so good

good , , so so good good , , you you see

see ! ! so so I I will will eat eat them

them in in a a box box . . and and I I will

will eat eat them them with with a a fox

fox . . and and I I will will eat eat them

them in in a a house house . . and and I

I will will eat eat them them with with a

a mouse mouse . . and and I I will will eat

will eat	I will	and I	• and	mouse	• a mouse
with a	them with	eat them	will eat	I will	
and I	• and	house	• a	house in	a in
eat them	will eat	I will	• and	I will	fox
a fox	with a	them with	eat them	will eat	
I will	and I	• and	box	• a box	in a
eat them	will eat	I will	so I	! so	see !
you see	, you	good	so good	, so	good
so good	are so	they are	• they	tree	• a tree
in a	and in	• and	a car	in a	and in

eat them them here here and and there

there . . say say ! ! I I will will eat

eat them them anywhere anywhere ! ! I I do

do so so like like green green eggs eggs and

and ham ham ! ! Thank Thank you you !

! Thank Thank you you , , Sam Sam I I am

am .

and there here and them here eat them

will eat I will I say I say there

I do I anywhere ! them anywhere eat them

eggs and green eggs like green so like do so

you ! Thank you ! Thank ham ! and ham

I am I , Sam I , Sam you , Thank you ! Thank

am